

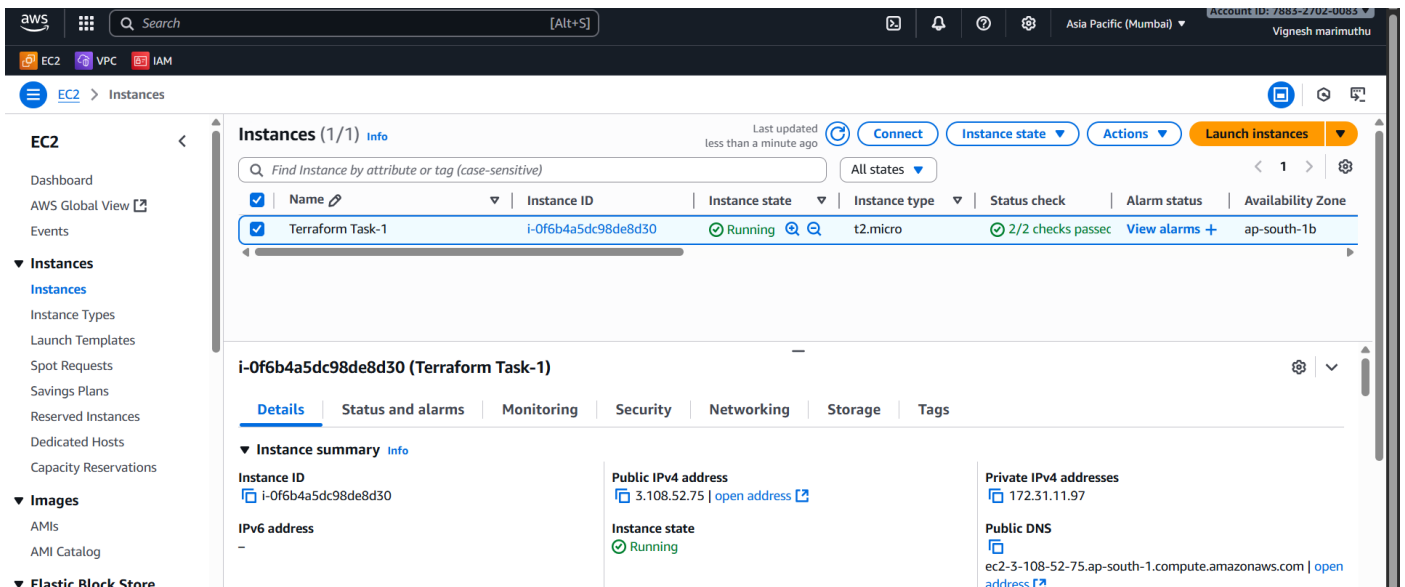
Terraform Task

Task Description:

Launch Linux EC2 instances in two regions using a single Terraform file.

Solution:

✓ 1. Launched EC2 and Installed Terraform



```
ubuntu@ip-172-31-11-97:~$ sudo apt update
sudo apt install -y wget unzip
wget https://releases.hashicorp.com/terraform/1.9.5/terraform_1.9.5_linux_amd64.zip
unzip terraform_1.9.5_linux_amd64.zip
sudo mv terraform /usr/local/bin/
terraform -version
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
```

i-0f6b4a5dc98de8d30 (Terraform Task-1)

PublicIPs: 3.108.52.75 PrivateIPs: 172.31.11.97

```
Your version of Terraform is out of date! The latest version
is 1.13.3. You can update by downloading from https://www.terraform.io/downloads.html
ubuntu@ip-172-31-11-97:~$
```

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- We first need Terraform on our EC2 Ubuntu machine. Terraform is a tool that lets us create AWS resources using code.

✓ 2. Configure AWS Credentials

```
You can now run: /usr/local/bin/aws --version
ubuntu@ip-172-31-11-97:~$ aws --version
aws-cli/2.30.4 Python/3.13.7 Linux/x86_64.0-1011-aws exe/x86_64.ubuntu.24
ubuntu@ip-172-31-11-97:~$ aws configure
AWS Access Key ID [None]: AKIA3PC7TWY2VJBMRPB7
AWS Secret Access Key [None]: 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
Default region name [None]: ap-south-1
Default output format [None]: json
ubuntu@ip-172-31-11-97:~$
```

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- Terraform needs AWS credentials to connect to your account.
- Used AWS CLI to configure.

✓ 3. Create Project Folder

```
ubuntu@ip-172-31-11-97:~$ mkdir terraform-multiregion && cd terraform-multiregion
ubuntu@ip-172-31-11-97:~/terraform-multiregion$ nano main.tf
ubuntu@ip-172-31-11-97:~/terraform-multiregion$
```

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- We keep Terraform files inside a folder for organization.
- `mkdir terraform-multiregion cd terraform-multiregion`

✓ 4. Write Terraform File

```
provider "aws" {
  region = "us-east-1"
}

provider "aws" {
  alias   = "west"
  region = "us-west-2"
}

# Get latest Ubuntu 20.04 AMI in us-east-1
data "aws_ami" "east_ubuntu" {
  most_recent = true
  owners      = ["099720109477"] # Canonical
  filter {
    name   = "name"
    values = ["ubuntu/images/hvm-ssd/ubuntu-focal-20.04-amd64-server-*"]
  }
}

# Get latest Ubuntu 20.04 AMI in us-west-2
data "aws_ami" "west_ubuntu" {
  provider = aws.west
  most_recent = true
}
```

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```
# East region EC2 instance
resource "aws_instance" "east_vm" {
  ami      = data.aws_ami.east_ubuntu.id
  instance_type = "t2.micro"
  provider = aws
  tags = {
    Name = "east-ubuntu"
  }
}
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-I To Bracket M-Q Previous
^X Exit ^R Read File ^N Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo M-G Copy ^Q Where Was M-W Next

```
tags = {
  Name = "east-ubuntu"
}

# West region EC2 instance
resource "aws_instance" "west_vm" {
  ami      = data.aws_ami.west_ubuntu.id
  instance_type = "t2.micro"
  provider = aws.west
  tags = {
    Name = "west-ubuntu"
  }
}

# Output public IPs
output "public_ips" {
  value = {
    east = aws_instance.east_vm.public_ip
    west = aws_instance.west_vm.public_ip
  }
}
```

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- Now create a file called main.tf with the following code.
- This file contains everything: two providers, two EC2 instances, and outputs.

✓ 5. Initialize Terraform

```
ubuntu@ip-172-31-11-97:~/terraform-multiregion$ terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.13.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
ubuntu@ip-172-31-11-97:~/terraform-multiregion$
```

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- We initialize Terraform to download required providers (AWS plugin).

✓ 6. Preview the Plan

```
ubuntu@ip-172-31-11-97:~/terraform-multiregion$ terraform plan
data.aws_ami.west_ubuntu: Reading...
data.aws_ami.east_ubuntu: Reading...
data.aws_ami.west_ubuntu: Read complete after 1s [id=ami-0a15226b1f7f23580]
data.aws_ami.east_ubuntu: Read complete after 2s [id=ami-0fb0b230890ccdie6]
aws_instance.east_vm: Refreshing state... [id=i-00c0eb3b4f219cd8f]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create
/+ destroy and then create replacement

Terraform will perform the following actions:

# aws_instance.east_vm must be replaced
/+ resource "aws_instance" "east_vm" {
  ~ ami                  = "ami-0c7217cdde317cfec" -> "ami-0fb0b230890ccdie6" # forces replacement
  ~ arn                  = "arn:aws:ec2:us-east-1:788327020083:instance/i-00c0eb3b4f219cd8f" -> (known after apply)
  ~ associate_public_ip_address = true -> (known after apply)
  ~ availability_zone       = "us-east-1a" -> (known after apply)
  ~ disable_api_stop        = false -> (known after apply)
  ~ disable_api_termination = false -> (known after apply)
}
```

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Plan: 2 to add, 0 to change, 1 to destroy.

Changes to Outputs:

```
+ public_ips = {
  + east = (known after apply)
  + west = (known after apply)
}
```

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

```
ubuntu@ip-172-31-11-97:~/terraform-multiregion$
```

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- This shows what Terraform will create (2 instances).

✓ 7. Apply and Create Resources

```
ubuntu@ip-172-31-11-97:~/terraform-multiregion$ terraform apply
data.aws_ami.west_ubuntu: Reading...
data.aws_ami.east_ubuntu: Reading...
data.aws_ami.west_ubuntu: Read complete after 1s [id=ami-0a15226b1f7f23580]
data.aws_ami.east_ubuntu: Read complete after 2s [id=ami-0fb0b230890ccd1e6]
aws_instance.east_vm: Refreshing state... [id=i-00c0eb3b4f219cd8f]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create
-/ destroy and then create replacement

Terraform will perform the following actions:

# aws_instance.east_vm must be replaced
/+ resource "aws_instance" "east_vm" {
  ~ ami               = "ami-0c7217cdde317cfec" -> "ami-0fb0b230890ccd1e6" # forces replacement
  ~ arn               = "arn:aws:ec2:us-east-1:788327020083:instance/i-00c0eb3b4f219cd8f" -> (known after apply)
  ~ associate_public_ip_address = true -> (known after apply)
  ~ availability_zone   = "us-east-1a" -> (known after apply)
  ~ disable_api_stop    = false -> (known after apply)
```

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```
Apply complete! Resources: 2 added, 0 changed, 1 destroyed.
```

```
Outputs:
```

```
public_ips = {
  "east" = "34.229.92.21"
  "west" = "35.88.180.3"
}
```

```
ubuntu@ip-172-31-11-97:~/terraform-multiregion$
```

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- This command actually launches the EC2 instances.

✓ 8. View Public IPs

```
ubuntu@ip-172-31-11-97:~/terraform-multiregion$ terraform output public_ips
{
  "east" = "34.229.92.21"
  "west" = "35.88.180.3"
}
ubuntu@ip-172-31-11-97:~/terraform-multiregion$
```

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- Terraform outputs the public IPs of both EC2 instances after creation.